

54LS42/DM54LS42/DM74LS42 BCD/Decimal Decoders

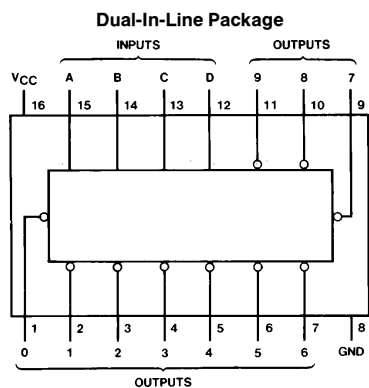
General Description

These BCD-to-decimal decoders consist of eight inverters and ten, four-input NAND gates. The inverters are connected in pairs to make BCD input data available for decoding by the NAND gates. Full decoding of input logic ensures that all outputs remain off for all invalid (10–15) input conditions.

Features

- Diode clamped inputs
- Also for applications as 4-line-to-16-line decoders; 3-line-to-8-line decoders
- All outputs are high for invalid input conditions
- Alternate Military/Aerospace device (54LS42) is available. Contact a National Semiconductor Sales Office/Distributor for specifications.

Connection Diagram



TL/F/6365-1

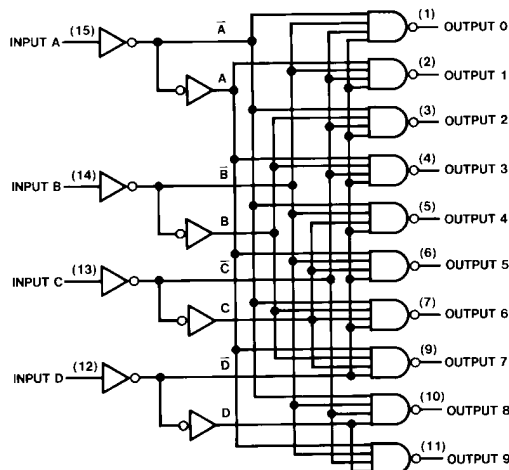
Order Number 54LS42DMQB, 54LS42FMQB,
DM54LS42J, DM54LS42W, DM74LS42M or DM74LS42N
See NS Package Number J16A, M16A, N16E or W16A

Function Table

No.	BCD Inputs				Decimal Outputs									
	D	C	B	A	0	1	2	3	4	5	6	7	8	9
0	L	L	L	L	L	H	H	H	H	H	H	H	H	H
1	L	L	L	H	H	L	H	H	H	H	H	H	H	H
2	L	L	H	L	H	H	L	H	H	H	H	H	H	H
3	L	L	H	H	H	H	H	L	H	H	H	H	H	H
4	L	H	L	L	H	H	H	H	L	H	H	H	H	H
5	L	H	L	H	H	H	H	H	H	L	H	H	H	H
6	L	H	H	L	H	H	H	H	H	H	L	H	H	H
7	L	H	H	H	H	H	H	H	H	H	H	L	H	H
8	H	L	L	L	H	H	H	H	H	H	H	H	L	H
9	H	L	L	H	H	H	H	H	H	H	H	H	H	L
Invalid	H	L	H	L	H	H	H	H	H	H	H	H	H	H
Invalid	H	L	H	H	H	H	H	H	H	H	H	H	H	H
Invalid	H	H	L	L	H	H	H	H	H	H	H	H	H	H
Invalid	H	H	L	H	H	H	H	H	H	H	H	H	H	H
Invalid	H	H	H	L	H	H	H	H	H	H	H	H	H	H
Invalid	H	H	H	H	H	H	H	H	H	H	H	H	H	H

H = High Level
L = Low Level

Logic Diagram



TL/F/6365-2

Absolute Maximum Ratings (Note)

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/Distributors for availability and specifications.

Supply Voltage	7V
Input Voltage	7V
Operating Free Air Temperature Range	
DM54LS and 54LS	−55°C to +125°C
DM74LS	0°C to +70°C
Storage Temperature Range	−65°C to +150°C

Note: The “Absolute Maximum Ratings” are those values beyond which the safety of the device cannot be guaranteed. The device should not be operated at these limits. The parametric values defined in the “Electrical Characteristics” table are not guaranteed at the absolute maximum ratings. The “Recommended Operating Conditions” table will define the conditions for actual device operation.

Recommended Operating Conditions

Symbol	Parameter	DM54LS42			DM74LS42			Units
		Min	Nom	Max	Min	Nom	Max	
V _{CC}	Supply Voltage	4.5	5	5.5	4.75	5	5.25	V
V _{IH}	High Level Input Voltage	2			2			V
V _{IL}	Low Level Input Voltage			0.7			0.8	V
I _{OH}	High Level Output Current			−0.4			−0.4	mA
I _{OL}	Low Level Output Current			4			8	mA
T _A	Free Air Operating Temperature	−55		125	0		70	°C

Electrical Characteristics over recommended operating free air temperature range (unless otherwise noted)

Symbol	Parameter	Conditions	Min	Typ (Note 1)	Max	Units
V _I	Input Clamp Voltage	V _{CC} = Min, I _I = −18 mA			−1.5	V
V _{OH}	High Level Output Voltage	V _{CC} = Min, I _{OH} = Max V _{IL} = Max, V _{IH} = Min	DM54	2.5	3.4	V
			DM74	2.7	3.4	
V _{OL}	Low Level Output Voltage	V _{CC} = Min, I _{OL} = Max V _{IL} = Max, V _{IH} = Min	DM54		0.25	V
			DM74		0.35	
		I _{OL} = 4 mA, V _{CC} = Min	DM74		0.25	
I _I	Input Current @ Max Input Voltage	V _{CC} = Max, V _I = 7V			0.1	mA
I _{IH}	High Level Input Current	V _{CC} = Max, V _I = 2.7V			20	μA
I _{IL}	Low Level Input Current	V _{CC} = Max, V _I = 0.4V			−0.4	mA
I _{OS}	Short Circuit Output Current	V _{CC} = Max (Note 2)	DM54	−20	−100	mA
			DM74	−20	−100	
I _{CC}	Supply Current	V _{CC} = Max (Note 3)		7	13	mA

Note 1: All typicals are at V_{CC} = 5V, T_A = 25°C.

Note 2: Not more than one output should be shorted at a time, and the duration should not exceed one second.

Note 3: I_{CC} is measured with all outputs open and all inputs grounded.

Switching Characteristics at $V_{CC} = 5V$ and $T_A = 25^\circ C$ (See Section 1 for Test Waveforms and Output Load)

Symbol	Parameter	From (Input) To (Output)	R _L = 2 kΩ				Units
			C _L = 15 pF		C _L = 50 pF		
			Min	Max	Min	Max	
t _{PHL}	Propagation Delay Time High to Low Level Output	A, B, C, or D (2 Levels of Logic) to Output		25		30	ns
t _{PHL}	Propagation Delay Time High to Low Level Output	A, B, C, or D (3 Levels of Logic) to Output		30		35	ns
t _{PLH}	Propagation Delay Time Low to High Level Output	A, B, C, or D (2 Levels of Logic) to Output		25		30	ns
t _{PLH}	Propagation Delay Time Low to High Level Output	A, B, C, or D (3 Levels of Logic) to Output		30		35	ns

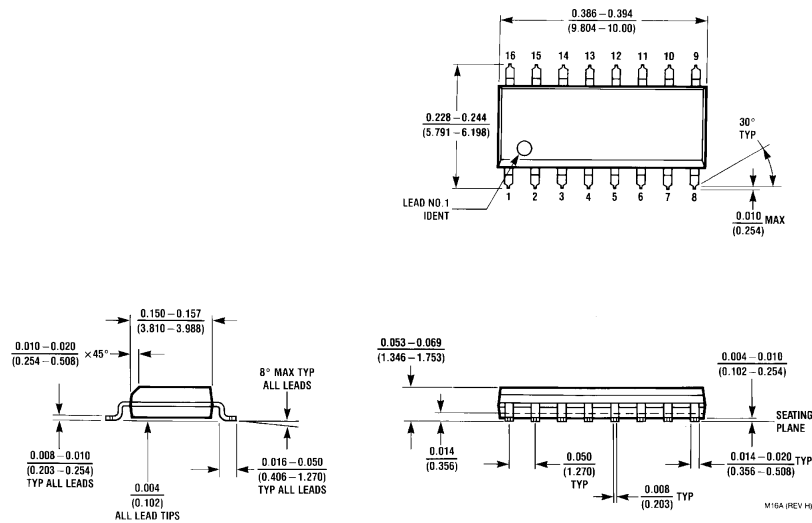
The drawing consists of three views of the J16A (REV L) component:

- Top View:** Shows a rectangular component with 16 pins. Dimensions include a total width of 0.785 [19.94] MAX, a pin pitch of 0.037 ± 0.005 [0.94 ± 0.13] TYP, and a pin diameter of 0.025 [0.64]. The component has a thickness of 0.220-0.310 [5.59-7.87] and a corner radius of R 0.005-0.020 [0.13-0.51] TYP.
- Side View:** Shows the component's profile with a height of 0.290-0.320 [7.37-8.13]. It features a glass sealant layer and a base with a thickness of 0.180 MAX [4.57]. The base has a width of 0.310-0.410 [7.87-10.41] and a 95° ± 5° TYP angle.
- End View:** Shows the component's end profile with a height of 0.200 [5.08] MAX TYP and a base thickness of 0.100 ± 0.010 [2.54 ± 0.25] TYP. The base has a width of 0.125-0.200 [3.18-5.08] TYP and a 90° ± 4° TYP angle.

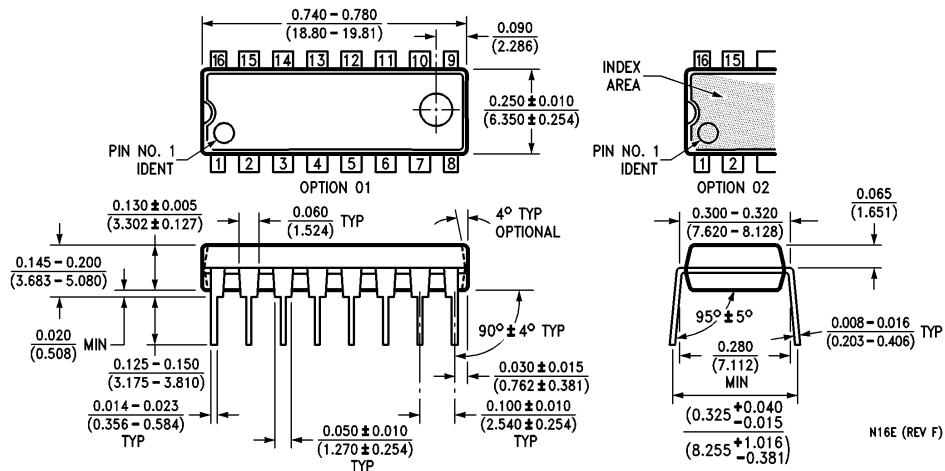
J16A (REV L)

J16A (REV L)

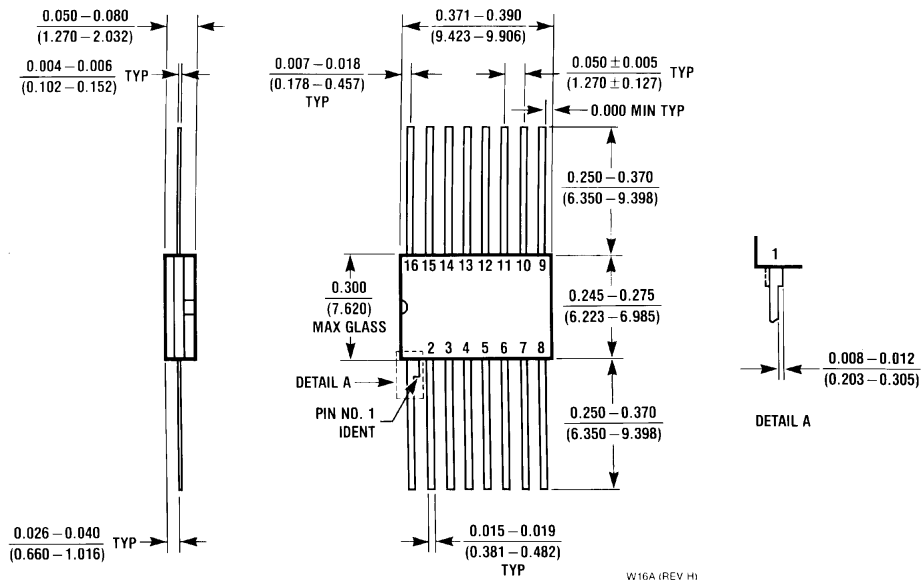
Physical Dimensions inches (millimeters) (Continued)



16-Lead Small Outline Molded Package (M)
Order Number DM74LS42M
NS Package Number M16A



16-Lead Molded Dual-In-Line Package (N)
Order Number DM74LS42N
NS Package Number N16E

Physical Dimensions inches (millimeters) (Continued)

16-Lead Ceramic Flat Package (W)
Order Number 54LS42FMQB or DM54LS42W
NS Package Number W16A

LIFE SUPPORT POLICY

NATIONAL'S PRODUCTS ARE NOT AUTHORIZED FOR USE AS CRITICAL COMPONENTS IN LIFE SUPPORT DEVICES OR SYSTEMS WITHOUT THE EXPRESS WRITTEN APPROVAL OF THE PRESIDENT OF NATIONAL SEMICONDUCTOR CORPORATION. As used herein:

1. Life support devices or systems are devices or systems which, (a) are intended for surgical implant into the body, or (b) support or sustain life, and whose failure to perform, when properly used in accordance with instructions for use provided in the labeling, can be reasonably expected to result in a significant injury to the user.
2. A critical component is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.



National Semiconductor Corporation
 1111 West Bardin Road
 Arlington, TX 76017
 Tel: 1(800) 272-9959
 Fax: 1(800) 737-7018

National Semiconductor Europe
 Fax: (+49) 0-180-530 85 86
 Email: cnjwge@tevm2.nsc.com
 Deutsch Tel: (+49) 0-180-530 85 85
 English Tel: (+49) 0-180-532 78 32
 Français Tel: (+49) 0-180-532 93 58
 Italiano Tel: (+49) 0-180-534 16 80

National Semiconductor Hong Kong Ltd.
 19th Floor, Straight Block,
 Ocean Centre, 5 Canton Rd.
 Tsimshatsui, Kowloon
 Hong Kong
 Tel: (852) 2737-1600
 Fax: (852) 2736-9960

National Semiconductor Japan Ltd.
 Tel: 81-043-299-2309
 Fax: 81-043-299-2408